

WEB ANALYTICS

Introduction to www

- The Web is a global set of documents, images and other resources, logically interrelated by hyperlinks and referenced with Uniform Resource Identifiers (URIs).
- URIs symbolically identify services, servers, and other databases, and the documents and resources that they can provide.
- Hypertext Transfer Protocol (HTTP) is the main access protocol of the World Wide Web. Web services also use HTTP to allow software systems to communicate in order to share and exchange business logic and data.

Definition and Scope

- According to the Web Analytics Association, the official definition of Web Analytics is: “The measurement, collection, analysis and reporting of Internet data for the purposes of understanding and optimizing Web usage”
- it is important to notice that Web Analytics is not a technology to produce reports; but rather it is a process that proposes a virtuous cycle for website optimization

What is web analytics?

- As a general term, web analytics means the analysis of the relationship between a website and the users of that website. However, in the field of web consultancy and e-commerce, web analytics has a field/industry specific meaning, which we will be covering in this section.
- When e-commerce began to emerge as a significant industry, it came with its own sets of needs, requiring professionals who could measure the effectiveness of strategies, the current state of affairs that a website was experiencing, what the demographics and demands of a website's users were.

What is web analytics?

- Most of all, as the internet grew, so did what we now call Big Data. To meet these new challenges and needs, a new field of expertise and knowledge evolved called Web Analytics.
- A major part of this new domain of information technology was the software used to measure and conduct web analytics. These genres of computer programmes and tools were instrumental in monitoring the relationship between websites and their users, the two poles in a dynamic relationship, which is the basis of web analytics.

The purpose of web analytics

- To give you a very basic summation of what kind of information a web analyst deals with, here are a few indicators or signals of a website that are meaningful and significant for web analytics:
- What is the gross quantity of visitors on a website or traffic quantum
- What is the number of unique visitors or visitors who are new
- The route involved in bringing different categories of traffic to the website. E.g., how many arrive through search engine results, how many get to the website through online marketing?
- What terms are trending on the search facility on the website, what are users searching for and in what quantity?

The purpose of web analytics

- What category of users searches for particular keywords or search terms?
- What is the average, minimum and maximum time being spent on the website by users?
- How many users are going beyond the main page, to deeper links?
- what are the links attracting the most second clicks beyond the homepage?
- What is the bounce rate? How many users arrive at the homepage and leave website?

Relevant Technology

- Internet & TCP/IP
- Client / Server Computing
- HTTP (HyperText Transfer Protocol)
- Server Log Files & Cookies
- Web Bugs

Server Log Files

- Website statistics are based on *server logs*. A server log is a simple text file which records activity on the server. There are several types of server log — website owners are especially interested in *access logs* which record hits and related information.
- Access logs come in several different formats but they all tend to look something like this:

```
213.60.233.243 - - [25/May/2004:00:17:09 +1200] "GET /internet/index.html HTTP/1.1" 200 8792 "http://www.mediacollege.com/video/streaming/http.htr
151.44.15.252 - - [25/May/2004:00:17:20 +1200] "GET /cgi-bin/forum/commentary.pl/noframes/read/209 HTTP/1.1" 200 8863 "http://search.virgilio.it/see
151.44.15.252 - - [25/May/2004:00:17:21 +1200] "GET /js/common.js HTTP/1.1" 200 2263 "http://www.mediacollege.com/cgi-bin/forum/commentary.pl/n
151.44.15.252 - - [25/May/2004:00:17:21 +1200] "GET /css/common.css HTTP/1.1" 200 8123 "http://www.mediacollege.com/cgi-bin/forum/commentary.
151.44.15.252 - - [25/May/2004:00:17:21 +1200] "GET /images/navigation/home1.gif HTTP/1.1" 200 2735 "http://www.mediacollege.com/cgi-bin/forum/c
151.44.15.252 - - [25/May/2004:00:17:21 +1200] "GET /data/zookeeper/ico-100.gif HTTP/1.1" 200 196 "http://www.mediacollege.com/cgi-bin/forum/com
151.44.15.252 - - [25/May/2004:00:17:22 +1200] "GET /adsense-alternate.html HTTP/1.1" 200 887 "http://www.mediacollege.com/cgi-bin/forum/commel
151.44.15.252 - - [25/May/2004:00:17:39 +1200] "GET /data/zookeeper/status.html HTTP/1.1" 200 4195 "http://www.mediacollege.com/cgi-bin/forum/co
```

Server Log Files

- Data saved generally includes:
 - IP address of the client
 - User name if logging on is necessary.
 - Date and Time
 - Page of file requested
 - Number of bytes returned,
 - Browser the client used
 - URL of page which contains the link
 - Any cookies.
 - Any errors

HISTORY OF WEB ANALYTICS

1990 – THE BIRTH OF THE WORLD WIDE WEB

- The Internet is essentially a constant dialogue of HTML code, flowing back and forth between a web user and a web server. On Christmas Day 1990, Sir Tim Berners-Lee successfully implemented the first dialogue of this kind, creating the Internet as we experience it today.

1993 – LOG FILES, CREATION OF WEBTRENDS

- Each time a certain HTML element is requested by a visitor, it is called a “hit” and is recorded into a log file. A hit may include text on a web page, an image, sound or a video file. However, during its modest beginnings, the Internet was comprised of mostly static pages limited to text and links. Therefore, when a page received a hit by a visitor, it was assumed that they were engaging with the entire contents of the page. As web usage increased, website owners became preoccupied with this metric.
- The development of log file analysis paved the path that led to the beginning of commercial web analytics, which was marked by the founding of WebTrends in 1993.

1995 – CREATION OF ANALOG

- Dr. Stephen Turner created Analog — the first log file analysis program to be completely free of charge. Until this time, web analytics could only be understood by tech teams, but Analog made the reports generated through log files more comprehensible to online business owners, with clear documentation and visual graphs. Finally, web analytics could be used by marketing professionals, as well.

1996 – HIT COUNTERS

- Web-Counter, the first widely used hosted hit counter service, was born. This sparked the trend of odometer-style hit counters being displayed on website landing pages.
- Accrue, Omniture, and WebSideStory were founded.

1997 – JAVASCRIPT TAGS

- As webpages began to include visuals and other elements besides text, it became clear that the number of hits a server accumulated no longer represented the number of pages requested.
- Javascript tagging became the new method of data collection to accurately report on diverse web traffic and trends. This is still the most widespread method of data collection today.

2004 – THE CREATION OF THE WEB ANALYTICS ASSOCIATION (WAA)

- Web analytics became more established as an essential tool for web optimization, providing increasingly complex solutions that reported massive amounts of data. The Web Analytics Association (WAA), now known as the Digital Analytics Association, was born.

2005 – GOOGLE BUYS URCHIN & LAUNCHES GOOGLE ANALYTICS

- Google Analytics quickly became the most widely used web analytics service on the market. Focusing heavily on quantitative analysis, it tied in directly with Google's other web marketing offerings.

2006 – THE BIRTH OF CLICKTALE (NOW CONTENTSQUARE) & IN-PAGE ANALYTICS

- The launch of In-Page analytics allowed website owners to see everything their visitors did on a webpage. Video session playbacks of visitor behavior delivered qualitative usability and conversion-based data, while heatmaps and form analytics provided online businesses with both quantitative statistics and qualitative behavioral data about website visitors.
- Clicktale (now Contentsquare) was established to offer solutions for businesses seeking conversion rate optimization, improved customer experience and increased user engagement through data sourced from analytics tools, such as mouse tracking.

2012 – UNIVERSAL ANALYTICS

- Google launched Universal Analytics, meaning that users could be tracked across multiple devices and platforms through user IDs. Beyond that, offline behavior began to be monitored, and customer data became richer with the addition of demographic and other information.
- Google Analytics for mobile apps was also launched, supplying app analytics via SDKs for Android and iOS. The world of web analytics grew to include the now standard mobile analytics.

2016 – MACHINE LEARNING ON MOBILE

- Google Analytics incorporated machine learning into its app analytics, giving marketers smarter insights on the go. This enabled a streamlined mobile Google Analytics experience, showing more relevant metrics, with real-time monitoring. This was a turning point in the analytics industry.

2017 – THE CLICKTALE (NOW CONTENTSQUARE) EXPERIENCE CLOUD

- By combining cognitive computing, machine learning and psychological research, the Clicktale (now Contentsquare) Experience Cloud was launched to solve critical digital questions for enterprises and enable them to provide top-quality customer experience. The new platform helps businesses better understand customer intent and behavior, and quantify impacts in order to more efficiently prioritize optimization tasks. This is made possible through the use of intuitive workflows and visualizations. The Clicktale (now Contentsquare) Experience Cloud pushes new frontiers, synthesizing complex behavioral patterns based on millisecond-level actions.

What's Next?

- Web and app design are heavily influenced by analytics: you wouldn't dream of designing your websites or apps without A/B testing and tweaking. Now, as the development of UX and UI is done hand in hand with derived insights from web analytics, this bond becomes ever stronger. The future is about optimizing this relationship as well as optimizing the insights accrued, in order to deliver maximum satisfaction to both customer and enterprise. Machine learning and cognitive computing are the focus of current web analytics evolution, and analytics providers continue to innovate in the field.
- We at Contentsquare are proud to be a part of this ever-evolving industry, and will continue to expand the role of web analytics in the years to come.



Web Server Statistics for **analog** documentation

Analysed requests from Sun-26-Apr-2015 08:26 to Sun-22-May-2016 23:44 (392.64 days).

General Summary

([Go To: Top](#) | [General Summary](#) | [Daily Report](#) | [Daily Summary](#) | [Hourly Summary](#) | [Status Code Report](#) | [File Size Report](#) | [File Type Report](#) | [Internal Search Query Report](#) | [Request Report](#))

This report contains overall statistics.

Figures in parentheses refer to the 7-day period ending 22-May-2016 23:45.

Successful requests: 191,709 (1,944)

Average successful requests per day: 488 (277)

Successful requests for pages: 17,697 (226)

Average successful requests for pages per day: 45 (32)

Failed requests: 3,463 (26)

Redirected requests: 321 (7)

Distinct files requested: 4,385 (1,097)

Distinct hosts served: 10,224 (178)

Corrupt logfile lines: 39

Data transferred: 2.13 gigabytes (16.55 megabytes)

Average data transferred per day: 5.56 megabytes (2.36 megabytes)

Daily Report

([Go To: Top](#) | [General Summary](#) | [Daily Report](#) | [Daily Summary](#) | [Hourly Summary](#) | [Status Code Report](#) | [File Size Report](#) | [File Type Report](#) | [Internal Search Query Report](#) | [Request Report](#))

Who Uses Web Analytics?

- **Web analytics** is essential for understanding user behavior and improving website performance.
- It provides insights that help professionals, from marketers to developers, optimize strategies, enhance user experience, and make data-driven decisions to achieve business goals.
- **Business Owners**
- **Digital Marketers**
- **Content Creators**
- **E-commerce Managers**
- **Web Developers**
- **SEO Specialists**
- **Product Managers**

- **Business Owners:**

Analyze customer behavior, monitor sales trends, and identify improvement areas. Web analytics helps them make informed decisions to optimize website performance, boost sales, and enhance customer satisfaction.

- **Digital Marketers:**

Measure campaign effectiveness, track conversion rates, and refine marketing strategies. Web analytics provides insights into audience behavior, enabling marketers to improve engagement and maximize ROI through data-driven adjustments.

- **Content Creators:**

Understand audience preferences, monitor content performance, and optimize content strategies. Web analytics helps creators produce relevant and engaging content that attracts and retains visitors, improving overall website engagement.

- **E-commerce Managers:**

Monitor product performance, analyze user behavior, and reduce cart abandonment. Web analytics identifies friction points in the purchasing process, helping managers enhance the user experience and increase conversions.

- **Web Developers:**

Identify technical issues, optimize website speed, and improve site functionality. Web analytics helps developers ensure a seamless user experience by addressing performance gaps and enhancing page load times.

- **SEO Specialists:**

Track organic traffic, analyze keyword performance, and refine SEO strategies. Web analytics provides insights to optimize content and improve search engine rankings, driving higher organic traffic.

- **Product Managers:**

Analyze user interactions, gather feedback, and enhance product offerings. Web analytics helps product managers align features with user needs, ensuring better product-market fit and higher customer satisfaction.

Categories of Web analytics:

On-site web analytics

- **Definition:**

On-site web analytics refers to the tracking and analysis of user behavior and interactions **on your own website or digital property**.

- **Purpose:**

To understand **how visitors use your site**, what content they engage with, and how effectively the site achieves its goals (e.g., conversions, downloads, sign-ups).

On-site web analytics

What It Measures:

- **Pageviews:** Number of times a page is loaded.
- **Sessions & Users:** How many visits and unique visitors you have.
- **Bounce Rate:** Percentage of users who leave after viewing only one page.
- **Conversion Rate:** Percentage of users who complete a goal (purchase, form submission).
- **Average Time on Page:** How long users spend on a specific page.
- **Exit Pages:** Where users most often leave the site.
- **Click Paths:** The route users take through your website.
- **Event Tracking:** Button clicks, form submissions, downloads, video plays, etc.
- **Custom Goals:** For example, adding a product to a cart or signing up for a newsletter

Benefits:

- Helps improve **user experience** by identifying UX problems.
- Aids in **conversion rate optimization (CRO)**.
- Informs **content strategy** by showing what content works best.
- Reveals **navigation patterns** and technical issues (slow pages, broken links).

Off-site web analytics

- **Definition:**

Off-site web analytics focuses on metrics and data that are **not directly collected from your own website**. Instead, it analyzes **your online presence in the broader digital ecosystem** — including your visibility, audience behavior, and reputation outside your site.

- **Purpose:**

To understand **how your brand performs externally**, how people find you, and what influences their decision to visit your site.

What It Measures:

- **Search Engine Rankings (SERP visibility):** Where your site appears in Google/Bing results.
- **Backlinks & Referral Traffic:** How many other websites link to yours, and how much traffic comes from them.
- **Social Media Metrics:** Likes, shares, comments, impressions, and engagement across platforms.
- **Mentions & Sentiment Analysis:** What people are saying about your brand or products online.
- **Competitor Insights:** Comparison of traffic, keywords, and marketing efforts.
- **Ad Performance:** Impressions, clicks, and conversions from external ad platforms.

Benefits:

- Provides insight into your **digital footprint** beyond your own website.
- Helps evaluate the **effectiveness of SEO** and content marketing efforts.
- Identifies **opportunities for partnerships or backlinks**.
- Monitors **brand reputation** and public perception.
- Supports **competitive analysis** for strategic decision-making.

Feature	On-site Web Analytics	Off-site Web Analytics
Data Source	Your own website	External platforms & third-party sources
Key Metrics	Pageviews, conversions, bounce rate	SEO rankings, social mentions, backlinks
Main Goal	Optimize user experience & conversion	Improve visibility, monitor brand performance
Tools	Google Analytics, Hotjar, Matomo	SEMrush, Ahrefs, Social media tools
Focus	Behavior of site visitors	How users find your site and talk about you

Web analytics activities

These are the core actions involved in the web analytics process that help businesses collect, analyze, and apply data insights effectively:

1. Data Collection

- Capturing raw data on user interactions using tools like Google Analytics or Adobe Analytics.
- Methods include JavaScript tracking codes, cookies, server logs, and tag management systems.
- Tracks pageviews, sessions, clicks, conversions, and more.

Data Collected Includes:

- Pageviews
- Time on site
- Clicks and scrolls
- Device, browser, and location
- Custom events (e.g., product shares)

2. Data Processing & Storage

- Cleaning and organizing raw data to ensure accuracy (e.g., filtering bots and spam).
- Session stitching (grouping user actions into sessions).
- Storing data securely for further analysis.

3. Reporting

- Turning raw data into readable reports and dashboards for analysis.
- Creating visual dashboards and reports to summarize key metrics.
- Reporting on user demographics, traffic sources, behaviors, and conversions.
- Sharing insights with stakeholders for decision-making.

4. Analysis

- Interpreting data to uncover patterns, trends, and insights.
- Performing segmentation (e.g., new vs. returning users, device types).
- Conducting funnel analysis and identifying drop-offs or bottlenecks.

5. Testing

- Implementing A/B or multivariate testing to experiment with website changes.
- Measuring impact on key metrics like conversion rate or bounce rate.

6. Optimization

- Applying insights and test results to improve website usability, content, and marketing effectiveness.
- Continuously iterating to enhance user experience and business outcomes

Key Components of Website Analytics

Website analytics refers to the collection, measurement, analysis, and reporting of data from a website to understand its performance and improve its user experience. The key components of a website analytics system include

- **Traffic Sources:** Understanding the sources of website traffic, such as organic search, referral, direct, social media, email, and advertising, is important for identifying the most effective marketing channels and optimizing for better performance.
- **User Behavior:** Measuring user behavior, such as page views, bounce rate, time on site, and conversion rate, helps to understand how users interact with the website and identify areas for improvement.

Key Components of Website Analytics

- **Content Analytics:** Analyzing the performance of individual pages, posts and other types of content is important for identifying the most popular and engaging content, as well as areas where users are dropping off.
- **Mobile Analytics:** With the growing use of mobile devices, it's important to understand how users interact with a website on mobile devices and optimize for a better mobile experience.
- **E-commerce Analytics:** For online stores, e-commerce analytics provides insights into sales, revenue, shopping cart abandonment, and product performance to help optimize for better conversion rates.

Key Components of Website Analytics

- **Custom Event Tracking:** Setting up custom event trackings, such as tracking button clicks, form submissions, and page scrolling, helps to understand user behavior and track specific actions on the website.
- **Integration with Other Tools:** Integrating website analytics with other tools, such as Google Ads, [Google Search Console](#), and CRM systems, provides a more comprehensive picture of website performance and user behavior.

Current Landscape and Challenges, Traditional Web analytics

Introduction:

Web analytics has become a crucial aspect of digital business strategies, enabling organizations to measure website performance, understand user behavior, and optimize digital experiences.

The rapid evolution of technology and regulatory frameworks has transformed the web analytics landscape, bringing new opportunities and challenges.

Current landscape of web analytics

AI and Machine Learning:

Integration Modern web analytics tools leverage artificial intelligence (AI) and machine learning (ML) to provide deeper insights into user behavior.

Predictive analytics and automated decision-making help businesses tailor user experiences dynamically.

Real-Time Data Processing:

The shift toward real-time analytics enables businesses to monitor user interactions instantly, allowing for quick optimizations and personalized marketing strategies.

Omnichannel Tracking Web analytics now extends beyond websites to track user behavior across multiple digital touchpoints, including social media, mobile apps, and smart devices, providing a holistic view of customer engagement.

Omnichannel Tracking:

Web analytics now extends beyond websites to track user behavior across multiple digital touchpoints, including social media, mobile apps, and smart devices, providing a holistic view of customer engagement.

Privacy and Compliance Regulations:

Stricter data protection laws such as GDPR, CCPA, and other regional regulations impact how businesses collect, store, and process user data.

Organizations must ensure compliance while maintaining analytical capabilities.

Server-Side Tracking:

With increasing restrictions on cookies, businesses are shifting towards server-side tracking to gather first-party data while improving privacy compliance and data security.

User-Centered Approach to Analytics:

web analytics should not just focus on numbers but also on improving the user experience (UX).

Metrics such as bounce rates, session duration, and task completion rates should be analyzed in the context of user behavior.

Defining Key Performance Indicators (KPIs):

The book highlights the importance of selecting KPIs that align with business and UX goals. Understanding user flows and drop-off points can help refine website design and navigation.

Segmentation for Deeper Insights:

Instead of relying solely on aggregate data, Beasley advocates for segmenting users based on behaviors, demographics, and acquisition sources.

This helps in tailoring experiences for different user groups.

Combining Qualitative and Quantitative Data:

Using qualitative research, such as usability testing and surveys, alongside quantitative analytics to gain a comprehensive understanding of user experience.

Iterative Optimization:

Web analytics should be an ongoing process, where insights drive iterative improvements.

A/B testing and usability studies help in refining UX design based on analytical findings.

Basic web analytics metrics

- Pageviews
- Bounce Rate
- Pages per Session
- Demographic Info
- Devices

1. Pageviews

- **Definition:**

A *pageview* is counted every time a user loads (or reloads) a page on your website.

Details:

- Each refresh or revisit to a page counts as a new pageview.
- If a user visits the same page multiple times during a session, each view is still counted separately.

Why it's important:

- Shows which pages are most visited.
- Helps identify content performance and user interest.
- **Example:**
If 1 user visits your homepage, then the blog, then the contact page, you get **3 pageviews**.

2. Bounce Rate

- **Definition:**

Bounce rate is the **percentage of visitors who leave the website after viewing only one page**, without taking any further action

Formula:

$$\text{Bounce Rate} = \left(\frac{\text{Single-page Sessions}}{\text{Total Sessions}} \right) \times 100$$

Why it's important:

High bounce rate may indicate: Irrelevant content, Poor user experience, Misleading traffic sources

However:

A high bounce rate isn't always bad. On single-page websites or blogs, users may get all they need on one page.

- **3. Pages per Session**

- **Definition:**

This metric shows the **average number of pages a user views during one session** on your site.

Formula:

$$\text{Pages per Session} = \frac{\text{Total Pageviews}}{\text{Total Sessions}}$$

Why it's important:

- Indicates user engagement.
- A higher number suggests visitors are exploring more content.

Example:

If 100 users generate 300 pageviews, pages/session = 3.

- **4. Demographic Info**

- **Definition:**

Demographic info refers to **user data such as age, gender, language, and location.**

- **Collected via tools like:**

Google Analytics (when enabled with advertising features)

- **Why it's important:**

- Helps tailor content, design, and marketing.
- Supports targeting in advertising.
- Reveals who your audience really is.

- **Example:**

If most users are aged 18–24 from India, you may adjust language tone, content topics, or ad targeting accordingly.

5. Devices

Definition:

This metric tracks **what kind of device users use to access your site**, such as:

- Desktop
- Mobile
- Tablet

Why it's important:

- Helps optimize user experience based on device.
- Influences design decisions (e.g., mobile responsiveness).

Device data helps you answer questions like:

- Do mobile users bounce more?
- Should I invest in a mobile app?
- Is my site performing worse on tablets?

Metric	Measures	Why It Matters
Pageviews	Number of times pages are viewed	Understand content popularity
Bounce Rate	% of users who leave after one page	Gauge content engagement and relevance
Pages per Session	Avg. pages viewed per visit	Measure engagement and navigation behavior
Demographic Info	Age, gender, language, location, etc.	Customize content and marketing strategy
Devices	Desktop vs. Mobile vs. Tablet usage	Guide UX design and performance optimization

Terminology in Web analytics

When implementing web analytics, a number of important terms and concepts need to be kept in mind. These include the following:

- Metrics

Metrics are indices or measurement standards used to gauge the performance of a website. Every web analysis is conducted by using a lot of different metrics, each of which measures and assess a different aspect of a website. On its own, the result of a metric may be indecisive. However, when put together, the results of multiple metrics can give a complete picture of web performance.

- Hits

Hits are also known as impressions. A hit or impression is registered when a user requests a file and your server delivers it. Such a file may be a word document, an image file, a download of content or even a file like JavaScript. If a page with four images is opened by a user, five hits will be registered on the server. Therefore, hits are recorded for every request, regardless of whether it results in a positive action from the user, like a purchase or a sale. For this reason, many web analyst experts assert that counting the number of hits is not one of the most reliable ways to gauge the performance of a website

- User sessions

A single user session is classified as the time the user took to browse a website in one go. By 'one go', it means that for the visit to count as a session, the activities performed have to be completed in a specified period of time. On average, one visit is determined by 30 clicks in 30 minutes.

- Page tracking

Page tracking is a very important metric because it tells you the number of views one particular page had in a predefined timeframe. This allows you to assess the performance of the content on that page.

- Page tagging

Page tagging helps web analysts determine the number of views a single page gets. For this metric, experts categorize a number of pages together in a group form to make it easy for them to be tracked.



Traffic Sources Overview

Traffic sources tell you **how people are finding your website**.

There are four main types:

- **Organic Traffic**
 - Visitors who come via unpaid search engine results (e.g., Google Search).
- **Social Traffic**
 - Visitors referred by social media platforms (e.g., Facebook, Twitter, LinkedIn).
- **Referral Traffic**
 - Visitors referred by links from other websites (excluding search engines or social media).
- **Direct Traffic**
 - Visitors who enter your site URL directly into the browser or have it bookmarked.

What Are Micro and Macro-Level Insights?

Micro-Level Insights = Small Details

- These focus on **individual user behavior** or small patterns.
- What page people landed on?
- How long they stayed?
- What device they used?
- What link they clicked?

Macro-Level Insights = Big Picture

- These look at **overall trends** from many visitors over time.
- Is traffic going up or down?
- Which source brings the most visitors?
- How many people bought something?
- Are you getting more traffic this month than last?

This is like looking at **all customers** over a year to see general trends.

refer notes and pdf

- Trinity
- Web analytics as a business process